Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 1 of 17 · Question 7 part b · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

7 (a) (i)	any two from: • pressure falls because of fall in volume; • fluid drains from capillaries; • and into lymph vessels • further away from heart ; • larger cross sectional area; • increased resistance/friction; • lumen of vein large(er)/artery narrow(er);		2
(ii)	• stay in capillary; • too large to pass out;		2
(b)	• plasma in test tube; • add biuret reagent; • (blue to) lilac/purple; • shows presence of protein;	ORA for last two points linked to mp3	4
(c)	• arterial end 4 - 3.3 = 0.7kPa; • (net) flow out; • venous end 1.6 - 3.3 = -1.7kPa; • (net) flow in;		4
(d)	• volume of blood decreases/less blood; • blood pressure drops;		2
			14 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 1 of 17 · Question 7 part b · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 2 of 17 · Question 1 part c · 3 marks

2020 JAN · Paper 01 · find part c in this mark scheme extract

Question number				
1	(a)	**The only correct answer is C** to slow down the growth of bacteria *A is incorrect because taste is personal preference* *B is incorrect because it does not kill the bacteria* *D is incorrect because the appearance is not affected*		1
	(b) (i)	protein		1
	(ii)	**The only correct answer is D** it does not contain energy from fat *A is incorrect because vitamins have no effect on body mass* *B is incorrect because cholesterol has no effect on body mass* *C is incorrect because it does contain energy from carbohydrates*		1
	(c)	• add Benedict's solution (to a sample of yogurt) (1) • heat gently (1) • brick red colour indicates the presence of sugar (1)	R if adding to glucose/reducing sugar Allow yellow, green, orange	3

Total for Question 1 = 6 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 3 of 17 · Question 6 part a.ii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

6 (a)	(i)	sucrase/invertase (1)		1
	(ii)	• enzyme is a protein (1) • biuret test (1) • purple if protein present/remains blue if absent (1)		3
	(iii)	• reusable (1) • stable (1)		2
(b)	(i)	• allows glucose molecules to pass (1) • prevents larger/other molecules/ blood cells from passing (1)		2
	(ii)	• breaks down/converts glucose (1) • into hydrogen peroxide (1) • which activates electrode (1) • more glucose results in more hydrogen peroxide (1)		4

Total 12 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 3 of 17 · Question 6 part a.ii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 4 of 17 · Question 7 part a.iii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 4 of 17 · Question 7 part a.iii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 5 of 17 · Question 3 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
3	(a) (i)	A (amino acids); B and D are components of lipids C is a carbohydrate		1
	(ii)	• add Biuret solution to test solution; • mauve/lilac, protein present; • remains blue, protein absent;		3
	(b) (i)	• suitable axes labels; • suitable scale; • correct plots; • suitable curves; • curves labelled;	max 3 for bar chart accept key	5
	(ii)	• denaturation; • loss of shape of active site/enzyme; • substrate no longer fits/decreases rate of activity;		3
	(iii)	any three from • B is much more active/works faster; • over whole temperature range; • higher optimum temperature for B/works fastest at 80°C; • active over a greater range of temperatures;		3

Total 15 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 5 of 17 · Question 3 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 6 of 17 · Question 2 part b · 3 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2 (a)	• $\frac{6.1}{100}$; • x85; • 5.2;	Allow ecf for mps 2 and 3 If 5.18 given for mp3 then max 2 marks Full marks for correct final answer	3
(b)	• add sample to boiling tube; • add biuret solution; • purple if present, (remains blue if absent);		3
(c)	• $\frac{100}{16}$; • = 6.25/7;	ecf 1 mark max for correct final answer to a division calculation	2
(d)	Any five from: • contains calcium; • contains vitamin D; • vitamin D allows calcium uptake/absorption (from intestine); • calcium absorbed/stored in bones/used to form compact bone; • to strengthen bones/increases bone density; • also contains protein which is required for (bone) development/growth;	Allow makes bones less brittle	5

Total 13

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 6 of 17 · Question 2 part b · 3 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 7 of 17 · Question 4 part b · 2 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

Question number				
4	(a) (i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
	(b)	• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 8 of 17 · Question 2 part b.iv · 5 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.iv in this mark scheme extract

2 (a)		• all components/all nutrients/all required nutrients/all food groups/protein, carbohydrates, fats, vitamins, minerals ; • in the correct quantity/correct proportion/ recommended intake;		2
(b) (i)		Protein/ /vitamins/named vitamin/minerals / named mineral/starch/fibre;		1
	(ii)	• $\frac{100}{4} = 25$; • 25×2.4; • = 60 g; **60 = 3 marks**	ECF for mp 2 and 3	3
	(iii)	• 4 slices = 470 (kJ)/1 slice = 118 (kJ); • $\frac{11\,800}{118} = 100.4$ slices; ***2 marks for 100.4 or 101or 100.43***	1 mark for 11800÷470 **or** 1 mark for 100	2
	(iv)	• break up/chop meat/ liquidise/small pieces; • place in test/boiling tube; • add alcohol/ethanol; • add water; • milky/cloudy/ milky white colouration shows presence of fat;	Apply the marking points independently Award mp3 and mp4 in any order. Accept mp3 and 4 in any order	5

Total = 13 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 8 of 17 · Question 2 part b.iv · 5 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 9 of 17 · Question 2 part c.i · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

2(a)	Any two from: carbohydrates/proteins/lipids/mineral salts (or named example)/fibre;;	Ignore named examples (1 mark for each two correct)	**(2)**

Question Number	Answer	Additional guidance	Mark
2(b)	Any two from: • lack of it causes scurvy; • (lack of vitamin C causes) damage to membranes/bleeding gums; • makes connective tissue/maintains/strengthens cell linings/gums/sticks/joins cells/tissues together;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)(i)	Any five from: • Add fresh juice/long-life juice (to a test tube); • add DCPIP drop by drop/slowly; • until blue colour disappears/solution turns colourless; • count number of drops/measure volume of DCPIP added; • repeat (to obtain mean); • compare volumes of DCPIP added;		**(5)**

Question Number	Answer	Additional guidance	Mark
2(c)(ii)	Any two from: • volume of juice; • temperature; • concentration DCPIP;	Allow amount for mp1. Ignore concentration	**(2)**

Total 11 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 9 of 17 · Question 2 part c.i · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 10 of 17 · Question 7 part b · 5 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7 (a) (i)	Any two from the following • causes cancer; • gene mutation; • incorrect protein made/shape of protein incorrect;		2
(ii)	• changing the way free radicals behave; • to allow normal flow of substances/named substance in and out of cells;		2
(iii)	• 40g = 60% of daily requirement; • $\frac{40 \times 100}{60}$; • = 67g;	Allow 66.7g	3
(b)	• add fresh orange juice to a test tube/named suitable container; • add DCPIP drop by drop until it no longer decolourises; • record the volume of DCPIP added; • repeat with the same volume of boiled orange juice; • compare volumes;		5
			Total 12

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 10 of 17 · Question 7 part b · 5 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 11 of 17 · Question 2 part a · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

2 (a)	4 of • add powder to water (in a test tube); • add Benedict's reagent/solution; • heat tube in a water bath; • observe any colour change/description of colour change; • wear goggles;		4 max
(b) (i)	powder Y;		1
(ii)	• powder W; • powder Z;		2
(iii)	powder X;		1
(iv)	protein;		1
			Total 9

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Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 11 of 17 · Question 2 part a · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 12 of 17 · Question 2 part a.i · 4 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
2	(a) (i)	• sample placed in test tube; • Benedict's solution added; • heat in a water bath; • blue to brick red if glucose present/stays blue if absent;		4
	(ii)	• sample from tubing placed in test tube/on spotting tile; • iodine solution added; • brown to blue-black if present/stays brown if absent;		3
	(iii)	• remove any starch/glucose; • spilt when putting solution into tubing; • to avoid contamination/doesn't affect result/investigation/ensure results are valid;		3
	(b)	• water in tube, positive for glucose but negative for starch • contents of tubing should test positive for starch and glucose; • Visking tubing has pores/partially permeable; • size(of pores) allows glucose to pass; • starch can't pass;		5

Total : 15 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 12 of 17 · Question 2 part a.i · 4 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 13 of 17 · Question 2 part a.ii · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
2	(a) (i)	• sample placed in test tube; • Benedict's solution added; • heat in a water bath; • blue to brick red if glucose present/stays blue if absent;		4
	(ii)	• sample from tubing placed in test tube/on spotting tile; • iodine solution added; • brown to blue-black if present/stays brown if absent;		3
	(iii)	• remove any starch/glucose; • spilt when putting solution into tubing; • to avoid contamination/doesn't affect result/investigation/ensure results are valid;		3
	(b)	• water in tube, positive for glucose but negative for starch • contents of tubing should test positive for starch and glucose; • Visking tubing has pores/partially permeable; • size(of pores) allows glucose to pass; • starch can't pass;		5

Total : 15 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 13 of 17 · Question 2 part a.ii · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 14 of 17 · Question 7 part b · 3 marks

2022 JAN · Paper 01 · find part b in this mark scheme extract

Question number					
7	(a)	(i)	carbon hydrogen and oxygen;		1
		(ii)	nitrogen	Allow sulfur	1
		(iii)	19.6 - 12; 7.6 g	Allow ECF	2
		(iv)	protein 1.2 g × 17 = 20.4 kJ; lipid 4.8 g × 37 = 177.6 kJ; total = 198 kJ;	Allow ECF for 2nd and 3rd mark	3
	(b)		add alcohol to fish; add water; cloudy/milky appearance/emulsion indicates lipid;		3

Total 10 marks

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Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 15 of 17 · Question 4 part a.i · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

4 (a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
(ii)	lipase		1
(iii)	fatty acids; glycerol;		2
(b) (i)	Any value between 8.6 and 8.8;		1
(ii)	independent = time/ temperature; dependent = pH;		2
(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 15 of 17 · Question 4 part a.i · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 16 of 17 · Question 4 part c · 4 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	Iodine (solution);		1
(b)	protein;		1
(c)	• add Benedict's/solution B to food in a test/boiling tube; • heat; • using a water bath; • wear goggles;		4
			Total 6

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 16 of 17 · Question 4 part c · 4 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 17 of 17 · Question 2 part b · 5 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number				
2	(a) (i)	• axes labelled with units; • axes correct way round and suitable scale; • points plotted correctly;; • straight lines joined; • lines identified;	time without with 0 65 110 30 120 215 60 130 230 90 95 200 120 80 190 150 65 180 180 65 170 don't include in published version	6
	(ii)	• without = 65 mg per 100cm^3; • with = 120 mg per 100cm^3;		2
	(iii)	any three from • without, level starts lower/level lower throughout; • smaller difference between highest and lowest values; • rises to lower levels/rise less quickly; • returns to normal (quicker)/same level 150-180;	ORA for person with diabetes	3
	(b)	• sample in a test/boiling tube; • add Benedict's reagent/solution; • heat in a water bath; • wear goggles; • brick red colour if glucose present; if candidate uses glucose testing sticks • sample in a test/boiling tube; • use of testing stick; • dip into urine; • compare colour with chart; • use of goggles;	Accept yellow/green/orange	5

Total 16 marks

Type MS · Describe how to carry out a food test, including DCPIP for vitamin C

Answer 17 of 17 · Question 2 part b · 5 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks